

Andrew Slattery

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🔗 AndrewSlattery

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Employment

Apr 2024 –	University of Cambridge <i>Research Associate</i> • Supervisor: Dr Jon Sterling • Funding provided by US Air Force Grant FA95502310728 ‘New spaces for denotational semantics’ • Extending the reach of denotational semantics with a homotopical and geometrical understanding of computation via higher dimensional category theory and topos theory
Oct 2024 –	Clare College, Cambridge <i>College Research Associate</i> • Concurrent College appointment, under which I undertake undergraduate supervision and admissions

Education

Oct 2020 – Mar 2024	University of Leeds <i>PhD in Category Theory</i> • Funding provided by EPSRC Scholarship • Supervisors: Dr Nicola Gambino, Dr Andrew Brooke-Taylor • Thesis Title: Commutativity of Relative Pseudomonads
Oct 2019 – Jun 2020	University of Cambridge <i>Masters of Mathematics (Part III) – Pass</i>
Oct 2016 – Jun 2019	University of Cambridge <i>Bachelor of Arts in Mathematics – First</i>

Research interests

My research is in higher category theory and its applications to logic and computer science. I am particularly interested in relative monads and pseudomonads, higher-dimensional multicategories, and pseudoalgebras, and in their use in contexts such as topos theory, fibred categories, and the categorical semantics of type theory.

Papers

2026	1. Slattery, A. & Sterling, J. Hofmann–Streicher lifting of fibred categories. <i>Logical Methods in Computer Science (conference version in LICS 2025)</i> (2026).
2025	2. Arkor, N., Saville, P. & Slattery, A. Bicategories of algebras for relative pseudomonads. <i>preprint, submitted</i> (2025).
	3. Arkor, N. & Slattery, A. Idempotence for relative monads. <i>preprint, submitted</i> (2025).
2023	4. Slattery, A. Pseudocommutativity and lax idempotency for relative pseudomonads. <i>Theory and Applications of Categories</i> (2023).

Talks and Posters

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| 2025 | | Logic in Computer Science (LICS) 40 , Singapore
<i>Hofmann–Streicher lifting of fibred categories</i> |
| 2024 | | Category Theory 2024 , Santiago de Compostela
<i>Bicategories of algebras for relative pseudomonads</i> (poster) |
| 2023 | | University of Manchester Category Theory Seminar , Manchester
<i>Presheaf Algebras are Cocomplete Categories</i>
Category Theory 2023 , Louvain-la-Neuve
<i>Pseudocommutativity and lax idempotency for relative pseudomonads</i> (poster)
107th Peripatetic Seminar on Sheaves and Logic (PSSL) , Athens
<i>Pseudocommutativity for relative pseudomonads</i> |
| 2022 | | Yorkshire and Midlands Category Theory Seminar 27 , Leeds
<i>Property-like structures and relative pseudomonads</i> |
| 2021 | | ‘Proofs, Constructions, Computations and Categories’ seminar , Leeds
<i>The 2-category of algebras over a relative 2-monad</i>
Postgraduate Logic Seminar , Leeds
<i>Well-quasiorders and Kruskal’s Tree Theorem</i> |

Teaching and Administrative Work

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| Oct 2024 – | | University of Cambridge
<i>College Research Associate</i> <ul style="list-style-type: none">• Supervised Mathematics Part IB Analysis and Topology (Michaelmas 2024) and Analysis II (Michaelmas 2025)• Supervised Mathematics Part II Graph Theory (Michaelmas 2024 and 2025)• Co-organised undergraduate admissions in Computer Science (December 2024 and December 2025) |
| Oct 2021 – Mar 2024 | | University of Leeds
<i>Teaching Assistant</i> <ul style="list-style-type: none">• MATH1055 Numbers and Vectors• MATH1050 Calculus and Mathematical Analysis• MATH1026 Sets, Sequences and Series• MATH1060 Introductory Linear Algebra• MATH1025 Number Systems <i>Co-organiser</i> <ul style="list-style-type: none">• Yorkshire and Midlands Category Theory Seminar• ‘Proofs, Constructions, Computations and Categories’ seminar |

Scholarships and Awards

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| Oct 2020 – Apr 2024 | | EPSRC PhD Scholarship |
| 2019 | | Harry Patten Prize for Best Result in Part II Mathematics at Clare |
| 2017 | | Amiya Batterji Prize for Best Result in Part I Mathematics at Clare |